

Mining Operations Knowledge Fabric

Comprehensive use case for ontology-driven fabric creation
Iron ore operations, autonomous fleet, processing, rail, port and green energy

TCS Weave - Hybrid Retrieval Intelligence Platform

Enterprise Mining Demo | July 2026

Upload this document to Weave to instantiate a Super Knowledge Fabric with discovered ontology, canonical knowledge graph, LLM executive insights, and agent-ready grounded retrieval.

1. Executive Summary

Large-scale integrated iron ore mining combines open-pit extraction, autonomous haulage, ore processing, heavy-haul rail, and port export into a single operating system. Many producers also run parallel decarbonisation and renewable energy programs. Operational data spans SCADA historians, fleet management systems (FMS), maintenance (SAP PM), safety (ICAM), environmental monitoring, production planning, and engineering document repositories.

This guide defines a Mining Operations domain model for TCS Weave Knowledge Fabric. When uploaded, Weave performs multi-stage fabric creation: document chunking and vector indexing, ontology discovery (entities, attributes, relationships), ontology enrichment via LLM (Bedrock/OpenAI), canonical knowledge graph build, exploratory co-occurrence graph, LLM executive insight briefing, and agent-ready retrieval APIs. The result is a Super Knowledge Fabric - not a simple chatbot index, but a governed semantic layer connecting pits, assets, production, maintenance, safety, and export logistics.

2. Business Challenge at Mining Enterprise Scale

- Siloed operational knowledge across mine sites, processing, rail, port, and energy programs.
- Engineers and planners spend hours reconciling equipment IDs, shift logs, maintenance history, and production KPIs.
- Autonomous fleet incidents require rapid cross-system reasoning (FMS + maintenance + geotechnical + safety).
- Regulatory and sustainability reporting demands traceable evidence from heterogeneous sources.
- AI copilots fail when retrieval is shallow - relationships between assets, routes, and production targets are lost.

Weave addresses this with Hybrid Retrieval Intelligence: classic RAG speed plus graph-aware ontology enrichment and secure APIs for enterprise agents.

3. Weave Super Knowledge Fabric - What Mining Operations Get

Capability	Outcome for Mining Ops
Knowledge Fabric	Domain-scoped Mining Ops fabric with guardrails (classif
Ontology Discovery	Auto-extracted entities: MineSite, HaulTruck, Crusher, T
Ontology Enrichment	LLM maps field synonyms, business definitions, and gover
Canonical Graph	Approved entity-relationship graph for schema-aware retr
Exploratory Graph	Co-occurrence graph from documents for discovery worksho
Knowledge Graph UI	Interactive D3 visualization with LLM executive briefing
Grounded Retrieval	Evidence-first context for shift handover, RCA, and plan
Test with LLM	Natural-language Q&A over fabric with Bedrock Claude or
Agent APIs	Partner/agent integration via secured REST endpoints
Composite Fabrics	Merge mine + rail + port PDFs/CSVs into unified Super KF

4. Mining Domain Scope

4.1 Physical Operations (Iron Ore Value Chain)

- Mine hubs: North Ridge, Central Range, Western Plateau, Iron Bridge Hub (reference labels).
- Open pit mining: drill and blast, load and haul, grade control, dewatering.
- Processing: primary/secondary crushers, ore processing facility (OPF), stockpiles, blending.
- Rail: autonomous and conventional ore trains, track maintenance, cycle time KPIs.
- Port: train unloaders, stockyards, shiploaders, vessel scheduling, export tonnes.

4.2 Autonomous and Digital Systems

- Autonomous Haulage System (AHS): haul trucks, health events, intervention logs, geofencing.
- Fleet Management System (FMS): dispatch, queue times, payload, fuel, cycle segments.
- Condition monitoring: vibration, temperature, oil analysis, predictive maintenance scores.

4.3 Sustainability and Green Energy

- Green hydrogen projects, renewable generation assets, scope 1/2/3 emissions tracking.
- Environmental approvals, water stewardship, rehabilitation plans, tailings governance.

5. End-to-End Operational Lifecycle

Stage 1 - Mine Planning: geological model, blast design, dig plan, fleet allocation, production target for shift (target_tonnes, target_fe_grade).

Stage 2 - Extraction: drilling program executed, blast fired, excavator loads haul truck, payload verified by onboard systems, cycle time recorded (load_queue_min, haul_min, dump_min).

Stage 3 - Processing: ore delivered to crusher, throughput tph recorded, downtime events logged (downtime_code, root_cause_category), product stockpile updated.

Stage 4 - Rail and Port: train consist assembled, rail cycle to port, stockpile blending, vessel load commenced, bill of lading reference assigned, export_confirmed_date.

Stage 5 - Maintenance and Reliability: work order raised (PM or breakdown), parts reserved, technician assigned, MTTR/MTBF updated, return to service.

Stage 6 - Safety and Governance: incident reported (ICAM), investigation status, corrective actions, regulatory notification if required, audit trail to governance_status.

6. Ontology Entity Catalog (Discovered and Enriched)

Weave Ontology Discovery scans this document and linked sources to propose the following entity types. Analysts review, approve, and version the canonical ontology before graph build.

Entity	Description	Key Identifier
MineSite	Operational mine or hub with pits and processing	site_id e.g. SITE-NR
OpenPit	Active pit or stage within a mine site	pit_id e.g. NR-NORTH-03
AutonomousHaulTruck	AHS-enabled haul unit	asset_id e.g. AHT-2147
Excavator	Loading unit (electric or diesel)	asset_id e.g. EX-8831
HaulCycle	Single load-haul-dump cycle event	cycle_id
Crusher	Primary/secondary crushing asset	crusher_id
ProcessingPlant	OPF or beneficiation facility	plant_id
Stockpile	Blended or product stockpile	stockpile_id
OreTrain	Rail consist moving ore to port	train_id / consist_id
RailSegment	Track segment with maintenance history	segment_id
PortTerminal	Export terminal facility	terminal_id
VesselShipment	Export shipment to customer	shipment_id / BL_ref
ProductionTarget	Shift/day production plan	target_id
SensorReading	Telemetry point-in-time	reading_id
MaintenanceWorkOrder	SAP PM or equivalent WO	wo_number
SparePart	Inventory part linked to WO	part_number
SafetyIncident	Safety/health/environment event	incident_id
PermitToWork	Controlled work authorization	ptw_id
EnvironmentalMonitor	Air/water/noise monitoring station	monitor_id
OreGradeSample	Assay / grade control sample	sample_id
HydrogenProject	Green energy initiative	project_id
Operator	Human operator or controller role	employee_id
ShiftLog	Shift handover narrative	shift_id

7. Ontology Relationship Model

Relationship	From -> To	Business meaning
CONTAINS	MineSite -> OpenPit	Site owns pit geometry and plan
OPERATES	Operator -> ShiftLog	Controller accountable for shift
ASSIGNED_TO	AutonomousHaulTruck -> HaulCycle	Truck executes cycle
LOADS_FROM	Excavator -> OpenPit	Material source for loading
TRANSPORTS_TO	HaulCycle -> Crusher	Ore movement path
FEEDS	Crusher -> ProcessingPlant	Plant feed stream
STORES_IN	ProcessingPlant -> Stockpile	Product placement
LOADS	OreTrain -> Stockpile	Rail loading event
DELIVERS_TO	OreTrain -> PortTerminal	Rail cycle completion
EXPORTS_VIA	VesselShipment -> PortTerminal	Ship loading
MONITORS	SensorReading -> AutonomousHaulTruck	Condition telemetry
RAISES	MaintenanceWorkOrder -> AutonomousHaulTruck	Asset maintenance
CONSUMES	MaintenanceWorkOrder -> SparePart	Parts usage
OCCURS_AT	SafetyIncident -> MineSite	Incident location
REQUIRES	PermitToWork -> MaintenanceWorkOrder	Safety control

SAMPLES	OreGradeSample -> OpenPit	Grade control loop
TARGETS	ProductionTarget -> MineSite	Plan vs actual
GOVERNS	EnvironmentalMonitor -> MineSite	Compliance monitoring
PART_OF	HydrogenProject -> MineSite	Decarbonisation linkage

8. Core Attribute Dictionary (Ontology Enrichment Targets)

Ontology Enrichment maps operational field names from SCADA, FMS, SAP, and PDFs to canonical attributes below.

Attribute	Type	Domain usage
target_tonnes	decimal	Shift production target (wet/dry tonnes)
target_fe_grade	decimal	Target iron grade %Fe
payload_tonnes	decimal	Actual truck payload
cycle_time_min	decimal	Total load-haul-dump minutes
availability_pct	decimal	Asset availability rolling 24h
utilisation_pct	decimal	Scheduled vs operating time
downtime_code	string	Standard downtime reason code
root_cause_category	enum	Equipment / Process / Weather / Safety
mttr_hours	decimal	Mean time to repair
mtbf_hours	decimal	Mean time between failures
fe_grade_pct	decimal	Assayed iron content
silica_pct	decimal	Impurity for blending
moisture_pct	decimal	Ore moisture for tonnage correction
train_cycle_hours	decimal	Mine-to-port rail cycle
vessel_deadweight	decimal	Ship capacity tonnes
export_tonnes	decimal	Final shipped quantity
incident_severity	enum	Near miss / LTI / environmental
icam_status	enum	Investigation workflow state
governance_status	enum	Logged / Approved / Escalated
sensitivity_level	enum	Public / Internal / Confidential / Restricted
data_classification	enum	Fabric guardrail classification
llm_reasoning_used	boolean	Agent used LLM for RCA or planning
confidence_band	enum	Low / Medium / High for automated insights

9. Detailed Scenario - Autonomous Haul Truck Intervention (AHT-2147)

On 2026-03-14 night shift at North Ridge Mine, Autonomous Haul Truck AHT-2147 recorded elevated brake temperature on SensorReading SR-991022 (value 118 deg C, threshold 105 deg C). FMS flagged health_event_code BRAKE_TEMP_HIGH. Fleet controller created ShiftLog SL-20260314-N2 noting reduced fleet speed in Pit NR-NORTH-03. MaintenanceWorkOrder WO-4509821 raised with priority 1; SparePart PN-BRK-7720 reserved. PermitToWork PTW-88341 required for ground intervention. MTTR target 4h; actual return_to_service 3.2h. ProductionTarget TG-20260314-NR reduced by 12,000 tonnes due to 2-truck fleet derate. SafetyIncident not raised (controlled stop). governance_status: Logged. sensitivity_level: Internal. LLM reasoning summary: Root cause likely brake pad wear; recommend inspection of entire AHT-214x series.

10. Detailed Scenario - Port Export and Blending

VesselShipment SHP-2026-8841 (customer: major steel mill) requires fe_grade_pct >= 62.0 and silica_pct <= 4.2. PortTerminal Export Berth A blends Stockpile STK-B1 (high Fe) with STK-B2 (medium Fe) to meet specification. OreTrain OT-4472 delivers 24,000 wet tonnes from Central Range rail cycle RC-9912 (cycle_time 18.4h). Shiploader SL-02 loads vessel hold 3; export_tonnes confirmed 178,400. Bill of lading BL-OPS-2026-8841 issued. EnvironmentalMonitor EM-PORT-07 records dust PM10 within limit. audit_id AUD-2026-4412 logged for export compliance.

11. How to Create the Mining Super Knowledge Fabric in Weave

Step 1 - Sign in and permissions

Login as admin or user with Create Knowledge, Ontology, and Fabrics features enabled.

Step 2 - Upload this PDF

Navigate to Create Knowledge -> Upload PDF. Name the fabric: Mining Operations. Add tags: mining, iron-ore, autonomous-fleet, rail, port. Set weave_domain: generic.

Step 3 - Wait for fabric build

Progress: ingest -> chunk -> embed -> index. document_count and total_chunks populate on completion.

Step 4 - Ontology Discovery

Open fabric -> Platform panel -> Discover Ontology. Weave extracts entities (Section 6) and relationships (Section 7) using LLM-assisted discovery. Review proposed elements in Ontology Workspace.

Step 5 - Review, enrich, approve

Ontology Enrichment recommends attribute mappings, synonyms (e.g. truck_id -> asset_id), and governance. Approve version -> Build Canonical Graph.

Step 6 - Explore Super KF

View Knowledge Graph: canonical + exploratory views, LLM Executive Insight briefing, entity analytics. Test with LLM using Bedrock Claude Sonnet 4.5 or OpenAI.

Step 7 - Optional composite

Add CSV exports (production actuals, maintenance WOs) or codebase fabric (FMS integration repo) and merge into composite Super KF for cross-source retrieval.

12. Composite Fabric Architecture for Mining

Recommended multi-fabric layout for enterprise rollout:

Fabric A - Operations Reference (this PDF + engineering standards).

Fabric B - Production and KPI CSV from data lake (Snowflake/Databricks connector).

Fabric C - Maintenance and SAP PM work order extracts.

Fabric D - Safety and ICAM investigation library.

Fabric E - Sustainability / green energy policy documents.

Fabric F - Codebase / integration APIs (optional, for agent developers).

Weave composite fabric merges retrieval across sources while preserving per-fabric guardrails.

13. Agent and Copilot Query Examples

- What is the end-to-end lifecycle from pit extraction to vessel export?
- Explain relationships between AutonomousHaulTruck, HaulCycle, and MaintenanceWorkOrder.
- What ontology attributes define production target vs actual for a shift?
- Describe the intervention workflow when brake temperature exceeds threshold on AHT-2147.
- How does Weave canonical graph differ from exploratory graph for mine planning?
- What governance fields apply to safety incidents and export compliance audits?
- Which entities are involved in port blending to meet fe_grade_pct specification?
- How should agents use grounded retrieval before invoking LLM for RCA?
- What is the recommended composite fabric strategy for multi-site mining operations?

- List downtime_code categories and root_cause_category values for processing plants.

14. Operational KPI Reference

KPI	Formula / definition	Typical owner
Availability	Operating hours / scheduled hours	Maintenance
Utilisation	Productive time / available time	Production
Cycle time	Load + haul + dump + queue	Mining
Payload compliance	Actual vs target payload %	Mining
Throughput (tph)	Tonnes processed per hour	Processing
Rail cycle time	Departure mine to port arrival	Rail
Stockpile variance	Model vs survey tonnes	Port
Fe grade variance	Assay vs target grade	Geology
MTTR	Repair duration mean	Maintenance
LTIFR	Lost time injuries per hours	Safety
Scope 1 intensity	tCO2e per tonne shipped	Sustainability

15. Governance, Guardrails and Security

- data_classification: Restricted for production telemetry; Confidential for commercial export data.
- pii_fields: operator employee_id masked in partner-facing APIs.
- compliance_tags: WHS, EPA, ISO45001, ISO14001 where applicable.
- encryption_at_rest / encryption_in_transit: enabled on fabric guardrails profile.
- approved_roles: mine_controller, maintenance_planner, port_logistics, safety_advisor.
- ontology version approval: no runtime graph mutation without approved ontology version.

16. Expected Super KF Outputs (Weave Platform)

Output	Description
Vector index	Semantic chunks from this guide + uploaded supplements
Ontology project	Versioned entity/attribute/relationship catalog
Canonical graph	Approved schema graph for deterministic retrieval
Exploratory graph	Co-occurrence graph for discovery sessions
LLM insight	Executive briefing: Executive Summary, Key Insights, Rec
Retrieval API	POST /query and /retrieve with evidence citations
Graph export	JSON export for downstream graph analytics
Migration JSON	If codebase fabric added - modernization blueprint packa

17. Mining Value Proposition with TCS Weave

Speed: First Super KF from this document in hours, not months of graph modeling.

Depth: Ontology + graph enrichment captures relationships plain RAG misses (truck -> WO -> pit -> target).

Control: Human-in-the-loop ontology approval, fabric guardrails, audit trails.

Flexibility: Bedrock on AWS for enterprise LLM; OpenAI for dev; agent APIs for mining copilots.

Scale: Composite fabrics across mine, rail, port, and energy programs; deploy on-prem, EC2, or AWS with IAM Bedrock.

Weave is the Hybrid Retrieval Intelligence Platform - Classic RAG core, graph-aware ontology enrichment, and secure agent-ready APIs - engineered for integrated mining and green energy operations.

are illustrative for ontology modeling.